

$$G = (N, T, P, S)$$

where $P: \alpha \rightarrow \beta$

if for context-free grammar: $\alpha \in N$ and $\beta \in (N \cup T)^+$

$$N \rightarrow (N \cup T)^+$$

Applications of Pumping Lemma

1. $\Sigma = \{0\}$

$$L_1 = \{0^{i^2} \mid i \geq 1\}$$

$$= \{0, 0000, 00000000, \dots\}$$

If this language is regular then we can write any string w of this language as,

$$w = xy^z$$

$$1 \leq \|xy\| \leq n \quad \text{as } \|y\| \neq 0 \quad \text{where } n \geq 1$$

xy^4 $n \leq \|xy^2\| \leq n^2 + n$, upper bound has increased so it's not in accordance with pumping lemma.

$$L_2 = \{0^{2^i} \mid i \geq 1\}$$

$$= \{0, 00, 0000, 000000, \dots\}$$

$$1 \leq \|xy\| \leq n$$

$$n \leq \|xy^4\| \leq 2n$$

$$\neq n \leq \|xy^{2 \cdot 2}\| \leq 2n$$

Substitution
Homomorphism } refer to book

L is
Prolog } look up

2. $\Sigma = \{0\}$, $\Delta = \{a, b\}$ what is $R_0 = ?$ s.t. $R_0 = L(R_0)$

$\hookrightarrow R_0 = 0^*$ is there $R_1 \subseteq R_0$?

Substitution: $f: \Sigma \rightarrow \Delta^*$

$$f(0) = aba$$

Then, for 00 we write

$$f(0)f(0) = abaaba$$

Is our regular expression closed under substitution?

$$w = x^i 0^j$$

$$i. f(\epsilon) = \epsilon$$

$$ii. f(x^i 0^j) = f(x^i) f(0^j)$$

$$\text{For, } f(000) = f(00) \cdot f(0)$$

$$= f(0) f(0) f(0)$$

$$= aba aba aba$$

$$= (aba)^3$$

Hence, for $w = 0^i$ then $(aba)^i$ $(aba)^i \neq a^i b^i a^i$

$$(aba)^+ = (a \cdot b \cdot a)^+$$

3. $f(0) = a$ and $f(1) = b^*$, then $f(0^+ 1^+)$ can give a regular expression. If L is the language $L = 0^+ (0+1)^+ 1^+$, then what would $f(L)$ be?

$$\hookrightarrow f(L) = f(0^+ (0+1)^+ 1^+) = \cancel{f(0^+)} \cdot \cancel{f(0+1)} \cdot \cancel{f(1^+)} \\ = \cancel{a^+}$$

Let, us take $0(0+1)1$

$$f(010) = f(0) f(1) f(0)$$

$$= 001 + 011$$

$$= a b^* a$$

$$\text{Let, } w_1 = 001$$

$$w_2 = 011$$

$$f(001) = f(0) f(0) f(1)$$

$$= a a b^*$$

$$f(011) = f(0) f(1) f(1)$$

$$= a b^* b^*$$

$$= a b^*$$

$$\text{Hence, } f(L) = f(001 + 011)$$

$$= \{ a a b^* + a b^* b^* \}$$

$$= \{ a (a + b^*) b^* \}$$

$$= a (a + b)$$

$$= a (a b^* + b^*)$$

$$f(L) = f(001 + 011)$$

$$= a a b^* + a b^* b^*$$

$$= a (a + b^*) b^*$$

= regular expression.

Hence, regular expression is closed under substitution.

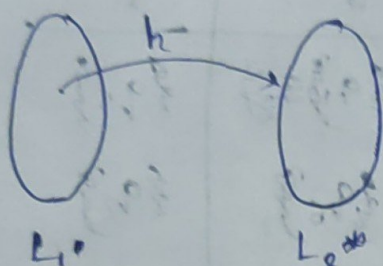
Decision Algorithm Theorem:-

The set of sentences accepted by a finite automata M with small end states is:

1. Non-empty iff the finite automata M $\|xy\| < n \rightarrow$ accepts a sentence of length $< n$,
- $\|xy\| = n \rightarrow$ 2. Infinite iff the automata accepts some sentences of length $(n \leq l \leq 2n)$.

3. $\mathbb{R}$

13/10/2023



Homomorphism $\rightarrow$ structure preserving mapping
Operation $\rightarrow$ Concatenation (in case of languages)

$$h(x_1, x_2) = h(x_1) * h(x_2)$$

- Regular sets are closed under homomorphism.

Myhill - Nerode Theorems

1. $L \subseteq \Sigma^*$ is accepted by some finite automata.
A set
2. L is the union of some of the equivalence classes of the a right invariant equivalence relation of finite index.
3. Let, equivalence relation R_L be defined by $x R_L y$ iff for all z in Σ^* , xz is an element of L when yz is in L .
Then, we can say that R_L is of finite index.

1 $\rightarrow$ 2 }
2 $\rightarrow$ 3 } Theorem is proved,
3 $\rightarrow$ 1 }

$$x, y \in L$$

$$\hat{\delta}(q_0, xz)$$

$$\hat{\delta}(\hat{\delta}(q_0, x), z) = \hat{\delta}(\hat{\delta}(q_0, y), z)$$

[Definition of right invariant]

$$x, y \in L$$

$$\hat{\delta}(q_0, x) = (q_i, x) \vdash$$

$$= q_i$$

$$\vdash (q_i, \epsilon)$$

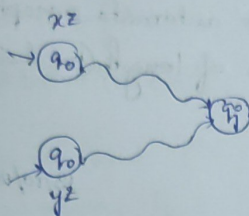
$$\hat{\delta}(q_0, y) = q_j \quad (q_0, y) \vdash$$

$$\vdash (q_j, \epsilon)$$

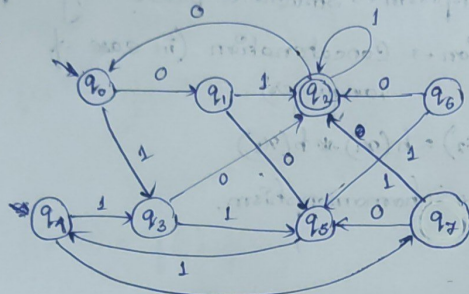
x and y are said to be invariant if $\hat{\delta}(q_0, xz) = \hat{\delta}(q_0, yz)$

$$\hat{\delta}(q_0, xz) \in F \quad z \in \Sigma^*$$

$$\hat{\delta}(q_0, yz) \in F \quad z \in L$$



Q.



q_6 is a dead state

01
10

$$\delta(q_0, 0) = q_1 \quad \delta(q_1, 0) = q_5$$

$$\delta(q_0, 1) = q_3 \quad \delta(q_1, 1) = q_2$$

$\delta(q_0, 01)$

$$\delta(q_0, 01) = q_2$$

$$\delta(q_1, 01) = q_4$$

$$\delta(q_0, 010)$$

δ	0	1
q_0	$\{q_1\}$	$\{q_3\}$
q_1	$\{q_5\}$	$\{q_2\}$
q_2	$\{q_0\}$	q_4
q_3	q_2	q_5
q_4	q_7	q_3
q_5	q_1	q_4
q_6	q_2	q_5
q_7	q_5	q_2

27th Oct, Mid sem TOC.

Closure: $L_1/L_2 = \{x \mid xy \in L_1 \text{ and } y \in L_2\}$

(20)

14/10/2023

 $x, y \in \Sigma^*$: indistinguishable $L' \subseteq \Sigma^*$: Regular Language/Set.where $\Sigma = \{a, b\}$ $\exists M' \mid L(M') = L'$ $M' = (Q, \Sigma, \delta, q_0, F)$ $\hat{\delta}(q_0, x)$ $\hat{\delta} : Q \times \Sigma^* \rightarrow Q$ $\delta : Q \times \Sigma \rightarrow Q$ $= q_i$ $\cong (q_0, x) \vdash (q_i, \epsilon)$ $\hat{\delta}(q_0, y)$ $= q_j$

$\begin{array}{c} \text{an} \\ \swarrow \quad \searrow \\ \text{Head} \quad \text{Tail} \\ \text{= Prefix of} \quad \text{= Postfix of} \\ \text{length } n \quad \text{= suffix} \end{array}$

 $w = xyz$

$\left. \begin{array}{l} x \\ xy \\ xyz \end{array} \right\} \text{ prefixes}$

$\left. \begin{array}{l} z \\ yz \\ xyz \end{array} \right\} \text{ pos suffixes}$

• Say, xz, yz both belong to L , then x and y are indistinguishable.

• If xz belong to one of xz, yz belongs to L and the other to L^c , then x and y are called distinguishable.

• R_L : indistinguishability relation or equivalence relation.

• If $xz \in L' \nmid$ then $yz \in L' \Rightarrow x R_L y$.

else, $xz \in \Sigma^* \setminus L'$ then $yz \in \Sigma^* \setminus L' \Rightarrow x R_L y$.

• Σ is the right invariance.

$(q_0, xz) = q_f$ where, $q_f, q_i \in F$.

$(q_0, x) = q_i$

as

i. $x R_L x$ ii. $x R_L y = y R_L x$ iii. $x R_L w$ and $x R_L y$ $\Rightarrow x R_L w$

Assignment: whether Prove that transitivity is possible in indistinguishable relation. i.e.,

If $x R_L y$ and $y R_L w$, then $x R_L w$.

• All the elements of L' need not be indistinguishable to x .

$L' = [x_1] \cup [x_2] \cup \dots \cup [x_k]$

• Equivalence classes are subsets of regular sets.

- No. of equivalence classes is finite.
But each equivalence class can be countably infinite.

Assignment: Equivalence classes are finite, find?

$M = (Q, \Sigma, \delta, q_0, F)$
equivalent

$p \equiv q$ iff $x \in \Sigma^*$ for each string in Σ^* ,

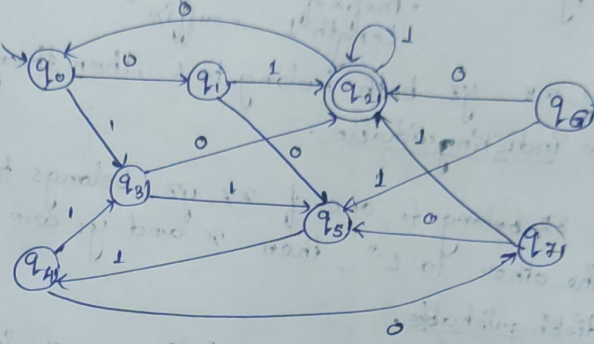
$\hat{\delta}(p, x) \in F$ iff $\hat{\delta}(q, x) \in F$

$\Sigma = \{0, 1\}$

$x = 0$, for (q_0, q_2)

$\hat{\delta}(q_0, 0) = q_1$

$\hat{\delta}(q_2, 0) = q_0$



$x = 1$

$\hat{\delta}(q_0, 1) = q_3 \rightarrow \text{Non-final}$

$\hat{\delta}(q_2, 1) = q_2 \rightarrow \text{final}$

Therefore, we can say (q_0, q_2) are not equivalent.

For (q_1, q_2)

$\hat{\delta}(q_1, 0) = q_3$ $\hat{\delta}(q_1, 1) = q_2$ $\hat{\delta}(q_2, 0) = q_0$ $\hat{\delta}(q_2, 1) = q_2$ $\therefore (q_1, q_2)$ are equivalent.

For (q_2, q_3)

$\hat{\delta}(q_2, 0) = q_0$ $\hat{\delta}(q_2, 1) = q_2$ $\hat{\delta}(q_3, 0) = q_4$ $\hat{\delta}(q_3, 1) = q_2$ $\therefore (q_2, q_3)$ are not equivalent.

(21)

For (q_1, q_0)

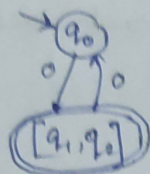
$$\hat{\delta}(q_1, 0) = q_1$$

$$\hat{\delta}(q_1, 1) = q_3$$

 $\therefore (q_1, q_3)$ are not equivalent.

$$\hat{\delta}(q_2, 0) = q_0$$

$$\hat{\delta}(q_2, 1) = q_2$$



- We will minimize the automata using this approach.

- We will a NFA.

- If a state is not reachable from q_0 are known to be dead state.

- All states from the dead state are eliminated from m' .

- $m' = (Q', \Sigma, \delta', q_0, F)$

- δ' will retain all the transitions except from the dead state.

1. Indistinguishability/Distinguishability is for strings. 18/10/2023

2. Equivalence of states. ^{regular set}

3. Given a $\Sigma, = \{a\}$ and let, $L \subseteq \Sigma^*$ accepted by a finite automata m .

Let, $x \in L(m)$ then $(p, x) \vdash$
 $\vdash (q, \epsilon)$

$(1, x) \vdash$
 $\vdash (6, \epsilon)$

4. $L_1 = 0^* 1^* 0^*$

$L_2 = 010^n, n \geq 0$

$0^n 10^n$ then 0's are equal on either sides.

$L_2 = (1+0)^*$

$\approx 010.0^*$

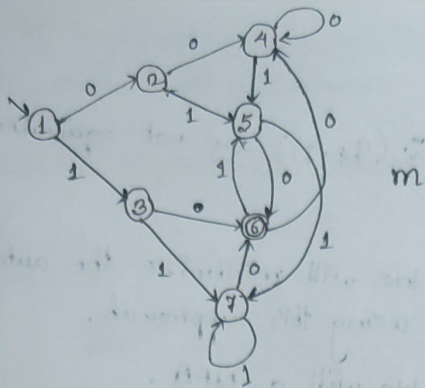
$L_3 = 1(1+0)^*$

We have to check if

$0^* 0^n 10^m \left[\begin{matrix} m, n \geq 0 \\ m, n \geq 1 \end{matrix} \right]$

$L_4 = 0^n 10^*, n \geq 0$

→ these are regular expression using pumping lemma. But these are not regular expressions.



$$\Sigma = \{0, 1\}$$

All the edges indicate a transition function.

Strings accepted by this automata are:

$$L(m) = \{10, \dots\}$$

$$x \in \Sigma^* = \{\epsilon, 0, 1, 00, 10, 01, 11, \dots\}$$

$$\left. \begin{array}{l} (5, x) \vdash (5, \epsilon) \\ (7, x) \vdash (7, \epsilon) \end{array} \right\} \text{ if } x = \epsilon \text{ for pair } (5, 7)$$

$$\text{if } x = 0$$

$$(5, x) \vdash 6$$

$$(7, x) \vdash 6$$

$$\text{if } x = 1$$

$$(5, x) \vdash 7$$

$$(7, x) \vdash 7$$

not equivalent

for pair (6, 7)

$$\text{if } x = \epsilon$$

$$(7, \epsilon) \vdash 7$$

$$(6, \epsilon) \vdash 6$$

$$\text{if } x = 0$$

for (7, 1)

$$(7, 0) \vdash 6$$

$$(6, 2)$$

$$(1, 0) \vdash 2$$

for (7, 2)

$$(7, 0) \vdash 6$$

$$(6, 4)$$

$$(2, 0) \vdash 4$$

for (7, 3)

$$(3, 0) \vdash 6$$

$$(6, 6)$$

for (7, 4)

$$(4, 0) \vdash 4$$

$$(6, 4)$$

for (7, 5)

$$(5, 0) \vdash 6$$

$$(6, 6)$$

$$\text{if } x = 0$$

for (5, 4)

$$(5, 4) \vdash (6, 4)$$

$$(5, 3) \vdash (6, 6)$$

$$(5, 2) \vdash (6, 4)$$

$$(5, 1) \vdash (6, 2)$$

	1	2	3	4	5	6	7
1	✓						
2		✓					
3	X	X	✓				
4				✓			
5	X	X	X	✓	✓		
6	X	X	X	X	X	✓	
7	X	X	✓	X			✓

Q9

if $x=0$ for $(4,3) \mapsto (4,6)$

$(4,2) \mapsto (4,4)$

$(4,1) \mapsto (4,2)$

if $x=0$ for $(2,1) \mapsto (6,2)$

$(3,2) \mapsto (6,4)$

if $x=0$ for $(2,1) \mapsto (4,2)$

$\delta(p,x) = \delta(q,x)$

$(p,q) \mapsto (r,r)$

$\delta(4,0) = \delta(2,0)$

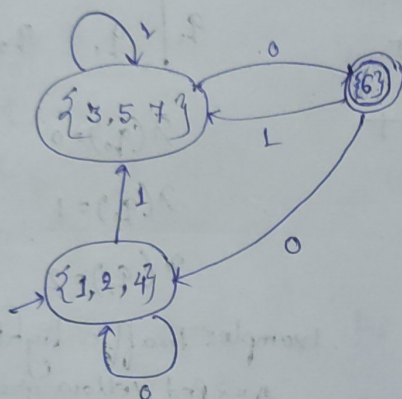
$(4,2) \mapsto (4,4)$

$\delta(4,1) = \delta(2,1)$

$(4,2) \mapsto (5,5)$

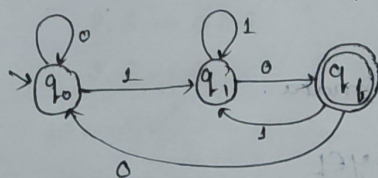
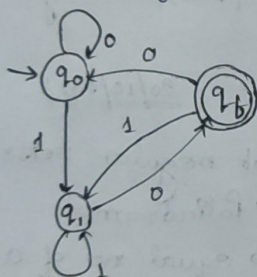
$\{2,4\}$

if $x=1$ for $(4,1) \mapsto (5,3)$



	0	1
(1,4)	(2,4)	(3,5)
(6,4)	(4,4)	(5,5)
(3,5)	(6,6)	(7,7)

	00	01
(1,4)	(4,4)	(7,7)

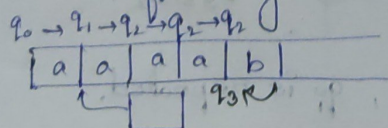


2DFA - 2 way Deterministic Finite Automata.

$\delta: (Q \times \Sigma) \rightarrow (Q \times \{L, R\})$

$\delta(q_0, a) = (q_1, R)$

$\{L, R\} \rightarrow$ movement of read head to left or right.



$(q_0, x) \mapsto (q_f, \epsilon)$ Accept

$(q_0, y) \mapsto (q_0, y')$

$(q_0, y') \mapsto (q_f, y'')$

q

$$\delta(q_0, a) = (q_1, R)$$

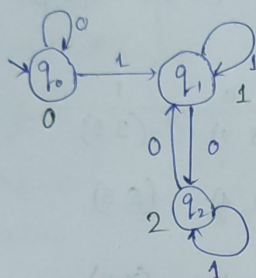
$$\delta(q_1, a) = (q_2, R)$$

$$\delta(q_2, a) = (q_2, R)$$

$$\delta(q_2, b) = (q_3, L)$$

$$\delta(q_3, a) = (q_1, R)$$

Moore Machine:



Outputs are associated with states, so each output state has an output.

$w = 00011010$

$(q_0, 00011010)$

$\Rightarrow 012111221$

(output)

$w = 10011010$

$(q_0, 10011010)$

$\Rightarrow 012111221$

Moore m/c

Mealy m/c

output alphabet)
 $(Q, \Sigma, \Delta, \delta, \lambda, q_0)$

$\delta: Q \times \Sigma \rightarrow Q$

$\lambda: Q \rightarrow \Delta$

$\Sigma = \{0, 1\}$

δ	0	1
q_0	q_0	q_1
q_1	q_2	q_1
q_2	q_1	q_2

$\lambda(q_0) = 0$

$\lambda(q_1) = 1$

$\lambda(q_2) = 2$

Example: Traffic light
 $\Delta = \{\text{red, yellow, green}\}$

20/10/2023

$m = a$ finite automata.

$|Q| = n$

$n > 0$

$\|w\| = n$

$\|xy\| \leq n$

$xyz \in L$

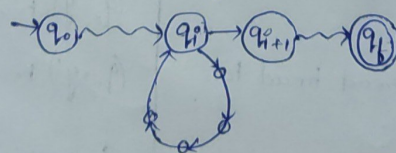
$xy^iz \in L$

$xy \in L$

$\|xy\| = n$ (trivial case)

If xy is accepted by m

$(q_0, xy) \rightsquigarrow (q_f, \epsilon)$



No. of edges = n

no. of vertices = $n+1$, q_0 is repeated.

Not regular sets:

↳ Palindromes

↳ equal no. of 0's and 1's.

$L = \text{palindrome}$ (proof by pumping lemma)

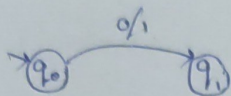
$w = \frac{1001001}{n} \frac{1}{y}$

$L_0, L_1 \subseteq P = \text{context free language}$

$L_3 \subseteq L_2$

(23)

Mealy Machine: Output is associated to the edge
 $m = (Q, \Sigma, \Delta, \delta, \gamma, q_0)$ $\delta: Q \times \Sigma \rightarrow Q$
 $\gamma: Q \rightarrow \Delta$

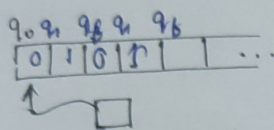


$\Delta = 0, 1, 2$

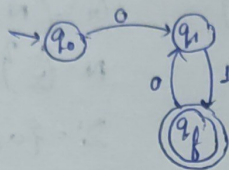
q	p	Δ
0	0	0
1	1	1
2	10	2
3	11	0
4	100	1

Example in book: To design a mealy m/c to output $p \bmod 3$.

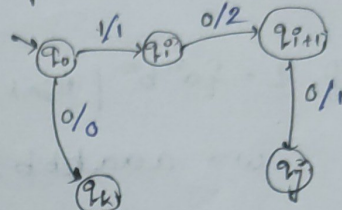
2-way automata:



$\delta(q_0, x) \neq (q_1, L)$
 (Totally forbidden)



$(01)^*(01)^*$



$G = (N, \Sigma, P, S)$

$P: \alpha \rightarrow \beta$

$S \xRightarrow{*} w \quad L(G)$

$\Sigma = \{0, 1\}$

$w = w_1 w_2$

$w_1 = 1101$

$S \xRightarrow{1} 150$

$\xRightarrow{1} 11500$

$\xRightarrow{2} 1105100$

$\xRightarrow{1} 110150100$

$\xRightarrow{0} 11010100$

$L_1 = \{w \mid w \text{ comprises of equal no of 0's and 1's and } w \neq \epsilon\}$

$P: 1. S \rightarrow 150$

$2. S \rightarrow 051$

$3. S \rightarrow \epsilon$

$w = w_1 w_1^R$

$w_1 = 1001$

$S \xRightarrow{1} 131$

$\xRightarrow{2} 10501$

$\xRightarrow{0}$

$\xRightarrow{3} 1001$

$w_1 = 1101$

$S \xRightarrow{1} 131$

$\xRightarrow{1} 11311$

$\xRightarrow{2} 1105011$

$L_2 = \{w \mid w \text{ is a palindrome of even length}\}$

$P: 1. S \rightarrow 150$

$2. S \rightarrow 051$

$1. S \rightarrow 151$

$2. S \rightarrow 050$

$3. S \rightarrow \epsilon$

$w = \frac{1001}{w_1} \frac{1001}{w_1^R}$

$w = \frac{1101}{w_1} \frac{1011}{w_1^R}$

$\xRightarrow{1} 110131011$

$\xRightarrow{3} 11011011$

$$Q. L = \{a^n b^n \mid n \geq 1\}$$

$$w = a a a b b b$$

$w \in L$

$$G = (N, \Sigma, P, S)$$

$$N = \{S\}$$

$$\Sigma = \{a, b\}$$

$$P: 1. S \rightarrow a S b$$

$$2. S \rightarrow ab$$

$$Q. L = \{a^n b^m \mid n \geq 1, m \geq n\}$$

$$w = a a a b b b b b$$

$$G = (N, \Sigma, P, S)$$

$$N = \{S\}$$

$$\Sigma = \{a, b\}$$

$$P: 1. S \rightarrow a S b$$

$$2. S \rightarrow b S$$

$$3. S \rightarrow b$$

$$Q. L = \{a^n b^{2n} \mid n \geq 1\} \quad w \mid w \text{ comprised of strings where } b's \text{ are twice the no. of } a's$$

$$w = b a b a a b b b b$$

$$G = (N, \Sigma, P, S)$$

$$N = \{S, B\}$$

$$\Sigma = \{a, b\}$$

$$P: 1. S \rightarrow \epsilon$$

$$2. S \rightarrow a S$$

$$3. S \rightarrow B S$$

$$4. B \rightarrow b b$$

$$5. S \rightarrow b S$$

$$6.$$

$$S \rightarrow a S b$$

$$S \rightarrow B S$$

$$S \rightarrow b S$$

$$S \rightarrow b a b S$$

$$S \rightarrow S b a b$$

$$S \rightarrow b S a b$$

$$S \rightarrow b a S b$$

$$S \xrightarrow{5} B S$$

$$\xrightarrow{2} b a S$$

$$\xrightarrow{3} b a b S$$

$$\xrightarrow{2} b a b a S$$

$$\xrightarrow{2} b a b a a S$$

$$\xrightarrow{3} b a b a a b S$$

$$\xrightarrow{3} b a b a a b b S$$

$$\xrightarrow{3} b a b a a b b b S$$

$$\xrightarrow{3} b a b a a b b b b S$$

$$\xrightarrow{1} b a b a a b b b b$$

no. of reductions

nd

by

see

(24)

Q. $w = vawb$, where $\|w\| = \|u\|$, $u, w \in (a+b)^*$

Q. $L = \{a^i u a^j \mid u \in (0+1)^*, \Sigma = \{0, 1, a\}^2\}$